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5.4.3. Particle Analysis - Practice Assessment

Analyse Particles

0 points possible (ungraded)

The Analyze Particles command in Fiji allows for ImageJ to measure different features of separate objects in a binary image. Moreover it has some built-in filters that can be applied in order to keep only a subset of elements in the results table

Which of the following actions can you perform within the Analyze Particles command?

Check all that apply

☒ You can filter the detected objects by size. ✓☒ You can display the results in a results table ✓☐ You can filter the detected objects by their intensity☒ You can exclude objects that are touching the edges ✓

i Answers are displayed within the problem

Set Measurements Command

0 points possible (ungraded)

The Analyze > Set Measurements command defines the columns that will be computed with the Analyze Particles command

What checkbox in Set Measurements adds the columns **Major, Minor, Angle**?

☐ Perimeter

☒ Fit ellipse ✓

☐ Bounding Rectangle

☐ Kurtosis

What column(s) appear when *Center of mass* is checked?

☐ X, Y

☐ BX, BY

☒ XM, YM ✓

☐ StdDe

Submit

i Answers are displayed within the problem

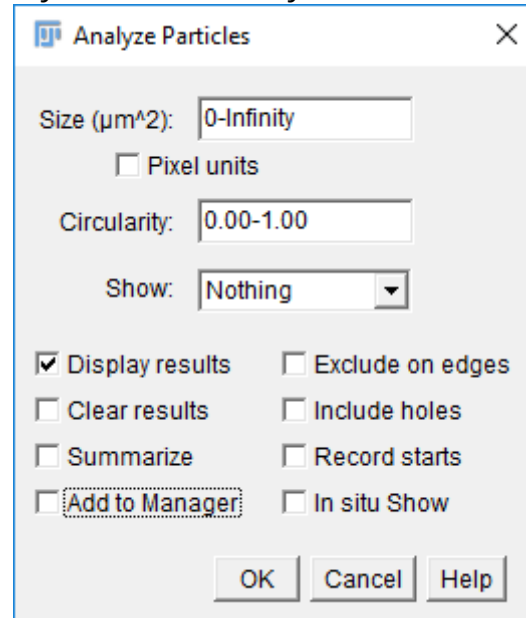
The Analyze Particles Command

0 points possible (ungraded)

Open the image 'Bacteria_Crop.tif', then copy and run the macro below

```
run("Set Measurements...", "area display redirect=None decimal=3");  
run("Duplicate...", "duplicate channels=2");  
run("Median...", "radius=2");  
setAutoThreshold("Otsu dark");  
setOption("BlackBackground", false);  
run("Convert to Mask");  
run("Watershed");
```

If you run the Analyze Particles command with the defaults as shown below, how many bacteria are detected?



✓ Answer: 27

Run Analyze Particles again with 'Exclude on edges' checked. How many bacteria do you have now?

✓ Answer: 24

By checking 'Summarize', you can get some summary statistics about the particles. What is the average size of the bacteria (Keep 'Exclude on edges' checked)?

✓ Answer: 1.306

What is the unit of the Area measurements for this image? (Check the image calibration in the metadata)

☐ um☒ um^2 ✓☐ pixels☐ pixels^2

i Answers are displayed within the problem

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